

DEPI – Software Testing Track

Part 01: Java Fundamentals

Instructor: Mina Younan

Session 09 – Task B

Question2: Create a Java Maven project to handle the following university scenario:

A university department offers different types of courses. There is an abstract class `Course` with a common attribute `name`. Two concrete classes inherit from it:

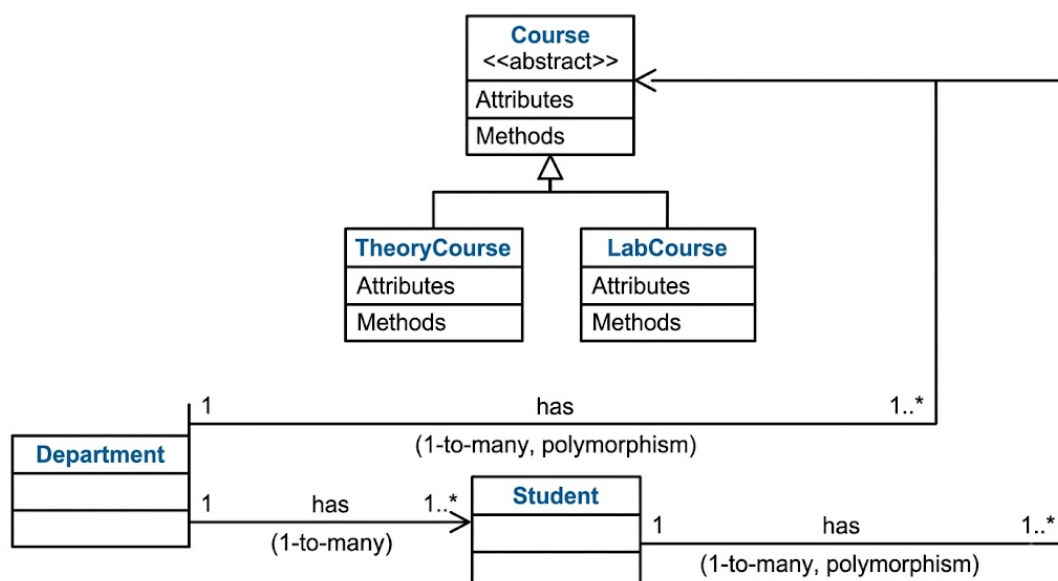
- `TheoryCourse`
- `LabCourse`

Each course type must implement a method that returns its type (“Theory” or “Lab”).

A `Student` can enroll in any kind of course. A `Department` can contain multiple students and multiple courses.

Requirements:

1. Use an **abstract class** for `Course`.
2. Apply **inheritance** (`TheoryCourse` and `LabCourse` extend `Course`).
3. Demonstrate **polymorphism** by storing both types of courses in a `List<Course>`.
4. Implement constructors with constructor chaining (`super(...)`).
5. Override the `toString()` method in the course classes.
6. Create a simple main class that:
 - Creates one department
 - Adds both Theory and Lab courses
 - Creates two students and enrolls them in different courses
 - Displays all courses and the courses of each student



Concept	Where
Abstract class	Course cannot be created directly
Inheritance	TheoryCourse and LabCourse extend Course
Polymorphism	List<Course> holds both types. toString() and getType() behave differently according to the real object
Constructor chain	super(name) in both child classes